# Monte Carlo Methods

## in Reinforcement Learning

*A comprehensive technical report covering Monte Carlo prediction, action-value estimation, control with and without exploring starts, and off-policy evaluation via importance sampling.*

**Subject:** Reinforcement Learning
**Topics:** MC Prediction, MC Control, Off-Policy Methods
**Level:** Advanced Undergraduate / Postgraduate
**Reference:** Sutton & Barto, *Reinforcement Learning* (2nd ed.)

Compiled for coursework assessment purposes

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# 1. Introduction

Reinforcement Learning (RL) is the study of how agents learn to make sequential decisions in order to maximise cumulative reward. A central challenge is *policy evaluation*: given a policy $\pi$, estimate the expected return from each state. Classical dynamic programming (DP) methods solve this efficiently when a perfect model of the environment is available. However, in most real-world applications the transition and reward functions are unknown, and model-based methods are infeasible.

**Monte Carlo (MC) methods** provide an elegant model-free alternative. They estimate value functions directly from *experience*—sequences of states, actions, and rewards sampled by interacting with the environment—without requiring any knowledge of the underlying Markov decision process (MDP). The core idea is simple: average the actual returns observed after visiting each state.

This report provides a rigorous yet intuitive treatment of the principal Monte Carlo techniques:

1. **Monte Carlo Prediction** – estimating $V_\pi$ from sampled returns.
2. **MC Estimation of Action Values** – why $Q_\pi(s, a)$ is necessary for control.
3. **Monte Carlo Control** – learning $\pi^*$ by alternating evaluation and improvement.
4. **MC Control without Exploring Starts** – $\varepsilon$-soft policies as a practical alternative.
5. **Off-Policy MC Prediction** – evaluating a target policy while following a different behaviour policy via importance sampling.

Throughout, we balance mathematical rigour with intuition, and include algorithms, diagrams, and worked examples.

# 2. Background: Episodic MDPs and Returns

## 2.1 The Reinforcement Learning Framework

We model the agent–environment interaction as a finite **Markov Decision Process** $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma)$ where $\mathcal{S}$ is the state space, $\mathcal{A}$ the action space, $\mathcal{P}$ the transition kernel, $\mathcal{R}$ the reward function, and $\gamma \in [0, 1)$ the discount factor.

A **policy** $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ maps states to distributions over actions. The agent's objective is to find a policy that maximises cumulative discounted reward.

## 2.2 Episodic Tasks

Monte Carlo methods apply naturally to **episodic tasks**, where the agent–environment interaction breaks naturally into finite episodes. Each episode begins at some start state $S_0$, follows the policy, and terminates at a terminal state $S_T$ after $T$ time steps.

An episode generates the trajectory:

$$S_0,\ A_0,\ R_1,\ S_1,\ A_1,\ R_2,\ \ldots,\ S_{T-1},\ A_{T-1},\ R_T,\ S_T.$$

### 2.3 The Return

**Definition: Discounted Return**

The **return** $G_t$ at time step $t$ is the total discounted reward from $t$ until the end of the episode:

$$G_t \;=\; \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1} \;=\; R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \cdots + \gamma^{T-t-1} R_T.$$

The discount factor $\gamma$ serves two purposes: it ensures convergence of the infinite-horizon sum and models the natural preference for immediate over delayed rewards. The return satisfies the **recursive Bellman identity**:

$$G_t = R_{t+1} + \gamma G_{t+1}, \tag{1}$$

which can be exploited for efficient computation during episodes.

### 2.4 Value Functions

The **state-value function** under policy $\pi$ is the expected return from state $s$:

$$V_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s], \qquad \forall s \in \mathcal{S}. \tag{2}$$

The **action-value function** conditions also on the chosen action:

$$Q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a], \qquad \forall s \in \mathcal{S},\ a \in \mathcal{A}. \tag{3}$$

## 3. Monte Carlo Prediction

### 3.1 Core Idea

The law of large numbers tells us that the sample mean of i.i.d. random variables converges to their expectation. Monte Carlo prediction exploits this: to estimate $V_\pi(s)$, we simply *average the returns* observed whenever the agent visits state $s$ across many episodes.

Formally, let $G^{(i)}(s)$ denote the return observed after the $i$-th visit to state $s$. Then:

$$V_\pi(s) \;\approx\; \widehat{V}(s) = \frac{1}{N(s)} \sum_{i=1}^{N(s)} G^{(i)}(s) \;\xrightarrow{N(s)\to\infty}\; V_\pi(s), \tag{4}$$

by the strong law of large numbers, provided returns are i.i.d. (guaranteed if episodes are independent).

### 3.2 First-Visit vs Every-Visit MC

A subtle but important distinction arises from how we handle multiple visits to the same state within a single episode.

**First-Visit Monte Carlo**

In **first-visit MC**, only the return following the *first* occurrence of state $s$ in each episode is used in the average. Subsequent visits within the same episode are ignored.

**Every-Visit Monte Carlo**

In **every-visit MC**, the return following *every* occurrence of state $s$—including repeated visits within the same episode—is included in the average.

Table 1: Comparison of First-Visit and Every-Visit MC prediction

| **Property** | **First-Visit** | **Every-Visit** |
|---|---|---|
| Bias | Unbiased estimator of $V_\pi(s)$ | Biased (for finite samples) |
| Consistency | Converges as $n \to \infty$ | Converges as $n \to \infty$ |
| Variance | Higher (fewer samples per episode) | Lower (more samples) |
| Theoretical | Cleaner analysis (i.i.d. returns) | Easier to implement |
| Usage | More common in theory | Common in practice |

Both methods converge to $V_\pi(s)$ as the number of episodes grows. Every-visit MC is often preferred in practice due to its simplicity, despite the bias for finite samples.

### 3.3 Algorithm: First-Visit MC Prediction

**Algorithm 1** First-Visit Monte Carlo Prediction for $V_\pi$

**Input:** policy $\pi$, discount $\gamma$
**Initialise:** $V(s) \leftarrow 0$ for all $s$; Returns$(s) \leftarrow \emptyset$ for all $s$
**repeat**
  Generate episode: $S_0, A_0, R_1, S_1, \ldots, S_{T-1}, A_{T-1}, R_T$ following $\pi$
  $G \leftarrow 0$
  **for** $t = T-1, T-2, \ldots, 0$ **do**
    $G \leftarrow \gamma G + R_{t+1}$ ▷ Compute $G_t$ using recursion (1)
    **if** $S_t \notin \{S_0, S_1, \ldots, S_{t-1}\}$ **then** ▷ First visit to $S_t$ in this episode
      Append $G$ to Returns$(S_t)$
      $V(S_t) \leftarrow$ average(Returns$(S_t)$)
    **end if**
  **end for**
**until** convergence

### 3.4 Incremental Updates

Maintaining a list of all observed returns is memory-intensive. An equivalent, more practical formulation uses an **incremental mean update**. After each episode, for each visited state $s$:

$$V(S_t) \leftarrow V(S_t) + \alpha\big[G_t - V(S_t)\big], \tag{5}$$

where $\alpha > 0$ is a step-size (learning rate). Setting $\alpha = 1/N(S_t)$ exactly recovers the running mean. Using a constant $\alpha$ gives a **weighted average** that emphasises more recent returns—useful in non-stationary environments.

**Intuition for (5):** The update moves $V(S_t)$ a fraction $\alpha$ of the way toward the observed return $G_t$. If $G_t > V(S_t)$, the state was better than expected, and $V$ is increased; if $G_t < V(S_t)$, the state was worse, and $V$ is decreased.

### 3.5 Comparison with Dynamic Programming

Table 2: Monte Carlo Prediction vs Dynamic Programming

| Criterion | MC Prediction | Dynamic Programming |
|---|---|---|
| Model required? | No (model-free) | Yes (full MDP model) |
| Bootstrap? | No (uses full return) | Yes (uses value estimates) |
| Updates | After each episode | After each sweep |
| Convergence | Stochastic, slower | Deterministic, faster |
| Memory | $O(\lvert\mathcal{S}\rvert)$ | $O(\lvert\mathcal{S}\rvert^2)$ for full DP |
| Applicable to | Episodic tasks | Episodic and continuing |

A key advantage of MC methods is their **locality**: the update for state $s$ does not depend on the values of other states, unlike DP where Bellman backups involve all successor states.

## 4. Monte Carlo Estimation of Action Values

### 4.1 Why Action Values Are Necessary for Control

A natural question is: why not use $V_\pi$ directly to improve the policy? The issue is that *without a model*, we cannot extract actions from the state-value function alone.

To perform **greedy policy improvement** with $V_\pi$, we need:

$$\pi'(s) = \arg\max_{a \in \mathcal{A}} \sum_{s'} \mathcal{P}(s' \mid s, a)\big[\mathcal{R}(s, a, s') + \gamma V_\pi(s')\big]. \tag{6}$$

This requires the transition model $\mathcal{P}$ and reward function $\mathcal{R}$—precisely what we do not have in model-free RL.

By contrast, with $Q_\pi$, policy improvement reduces to:

$$\pi'(s) = \arg\max_{a \in \mathcal{A}} Q_\pi(s, a), \tag{7}$$

which requires **no model at all**. This is the fundamental reason why MC control operates on $Q_\pi$ rather than $V_\pi$.

### 4.2 Estimating $Q_\pi$ from Episodes

Action values are estimated in exact analogy to state values. For each state-action pair $(s, a)$ visited in an episode, we record the subsequent return and average over visits:

$$Q_\pi(s, a) \;\approx\; \widehat{Q}(s, a) = \frac{1}{N(s, a)} \sum_{i=1}^{N(s,a)} G^{(i)}(s, a). \tag{8}$$

The same first-visit / every-visit distinction applies: first-visit MC for $Q$ records only the return following the first occurrence of $(s, a)$ in each episode.

### 4.3 Backup Diagram for Monte Carlo

Unlike DP, which backs up one step and uses estimated successor values, MC backs up the **entire trajectory** to the terminal state.

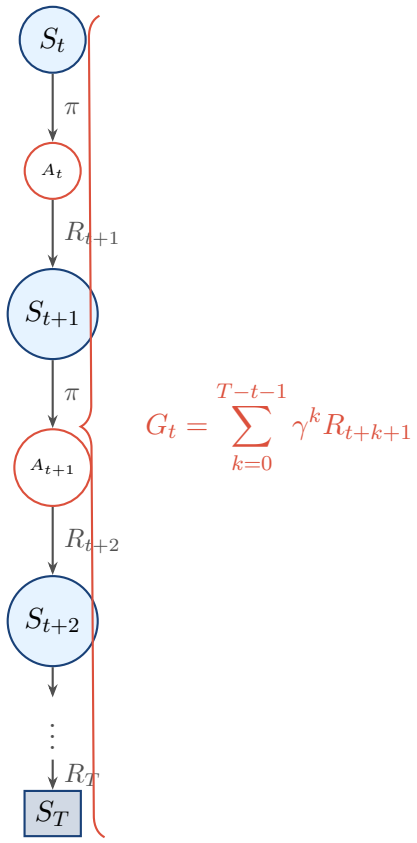



Figure 1: MC backup diagram. The update for $Q(S_t, A_t)$ uses the full observed return $G_t$ from the current state-action pair all the way to the terminal state $S_T$. There is no bootstrapping.

### 4.4 The Exploration Problem

A critical complication for action-value estimation is ensuring that **all state-action pairs are visited**. If the policy $\pi$ is deterministic, many $(s, a)$ pairs will never be visited, making it impossible to estimate $Q(s, a)$ for non-chosen actions. This **exploration** challenge is central to MC control.

## 5. Monte Carlo Control

### 5.1 Generalised Policy Iteration

MC control follows the **generalised policy iteration (GPI)** framework: alternate between

- **Policy Evaluation**: estimate $Q_\pi$ using MC methods.
- **Policy Improvement**: make the policy greedy with respect to the current $Q$ estimate.

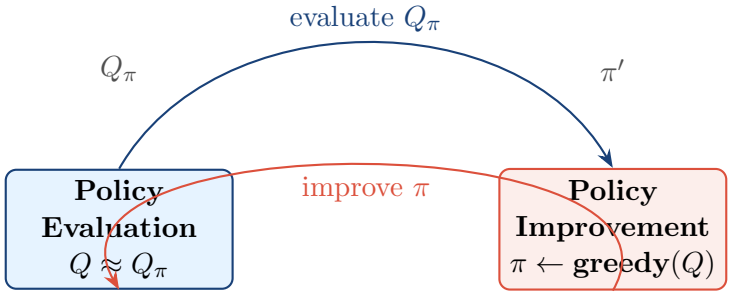



Figure 2: Generalised Policy Iteration. Policy evaluation and improvement alternate until convergence to the optimal policy $\pi^*$ and value function $Q^*$.

Under mild conditions, this process converges to the optimal policy $\pi^*$ and optimal action-value function $Q^*$.

### 5.2 MC-ES: Monte Carlo with Exploring Starts

The **exploring starts** assumption resolves the exploration problem by requiring that every state-action pair has a positive probability of being selected as the starting state of an episode:

$$P(S_0 = s,\ A_0 = a) > 0, \qquad \forall\, (s, a) \in \mathcal{S} \times \mathcal{A}.$$

This guarantees that, with sufficiently many episodes, every $(s, a)$ pair will be visited—and hence $Q(s, a)$ can be estimated for all pairs.

**Algorithm 2** Monte Carlo ES (Exploring Starts)

**Initialise:** $Q(s,a)$ arbitrarily; $\pi(s) \leftarrow \arg\max_a Q(s,a)$; $\text{Returns}(s,a) \leftarrow \emptyset$
**repeat**
  Choose $S_0 \in \mathcal{S}$, $A_0 \in \mathcal{A}$ with all pairs having $P > 0$ ▷ Exploring start
  Generate episode from $S_0, A_0$ following $\pi$
  $G \leftarrow 0$
  **for** $t = T{-}1, T{-}2, \ldots, 0$ **do**
    $G \leftarrow \gamma G + R_{t+1}$
    **if** $(S_t, A_t) \notin \{(S_0, A_0), \ldots, (S_{t-1}, A_{t-1})\}$ **then**
      Append $G$ to $\text{Returns}(S_t, A_t)$
      $Q(S_t, A_t) \leftarrow \text{average}(\text{Returns}(S_t, A_t))$
      $\pi(S_t) \leftarrow \arg\max_a Q(S_t, a)$ ▷ Greedy improvement
    **end if**
  **end for**
**until** convergence

### 5.3 Greedy Policy Improvement and Convergence

**Proposition 5.1** (Policy Improvement Theorem)**.** *Let $\pi$ be a stochastic policy and $\pi'(s) = \arg\max_a Q_\pi(s,a)$ be the greedy improvement. Then:*

$$Q_\pi(s, \pi'(s)) \geq V_\pi(s), \qquad \forall s \in \mathcal{S},$$

*which implies $V_{\pi'}(s) \geq V_\pi(s)$ for all $s$, with equality only when $\pi$ is already optimal.*

Convergence of MC-ES to $Q^* = Q_{\pi^*}$ can be established by noting that:

1. The exploring starts ensure all $(s,a)$ pairs are sampled infinitely often.
2. Each greedy improvement step cannot decrease expected return.
3. The process is equivalent to GPI, which is known to converge.

### 5.4 Why Exploring Starts Are Often Impractical

Despite its elegance, MC-ES has a critical limitation: **it is often impossible to specify arbitrary starting conditions** in real applications.

**Limitations of Exploring Starts**

1. **Physical systems:** A robot, autonomous vehicle, or game agent cannot arbitrarily reset to any state-action pair. The environment dictates the starting state.
2. **Simulated environments:** Even when a simulator exists, visiting rare or dangerous states (e.g., edge cases in autonomous driving) is computationally costly or hazardous.
3. **Continuous spaces:** In large or continuous state-action spaces, ensuring all pairs are visited with positive probability is practically infeasible.
4. **Real-world learning:** Online interactions with users, patients, or markets cannot be controlled to generate arbitrary starting conditions.

These limitations motivate the on-policy methods of Section 6.

## 6. MC Control without Exploring Starts

### 6.1 The Key Idea: Soft Policies

Instead of exploring starts, we ensure sufficient exploration by *never fully committing to a single action*. A policy is called $\varepsilon$**-soft** if it assigns positive probability to every action in every state:

$$\pi(a \mid s) \geq \frac{\varepsilon}{|\mathcal{A}|}, \qquad \forall s \in \mathcal{S},\ a \in \mathcal{A},\ \varepsilon > 0.$$

All $\varepsilon$-soft policies generate trajectories that visit all $(s,a)$ pairs infinitely often (with probability 1 as episodes accumulate), ensuring consistent estimation.

## 6.2 $\varepsilon$-Greedy Policies

The most common $\varepsilon$-soft policy is the **$\varepsilon$-greedy policy**:

$$\pi(a \mid s) = \begin{cases} 1 - \varepsilon + \dfrac{\varepsilon}{|\mathcal{A}|} & \text{if } a = \arg\max_{a'} Q(s, a'), \\ \dfrac{\varepsilon}{|\mathcal{A}|} & \text{otherwise.} \end{cases} \tag{9}$$

This policy acts **greedily** with probability $1 - \varepsilon$ (exploiting current knowledge) and **uniformly at random** with probability $\varepsilon$ (exploring). The parameter $\varepsilon$ controls the exploration-exploitation trade-off.

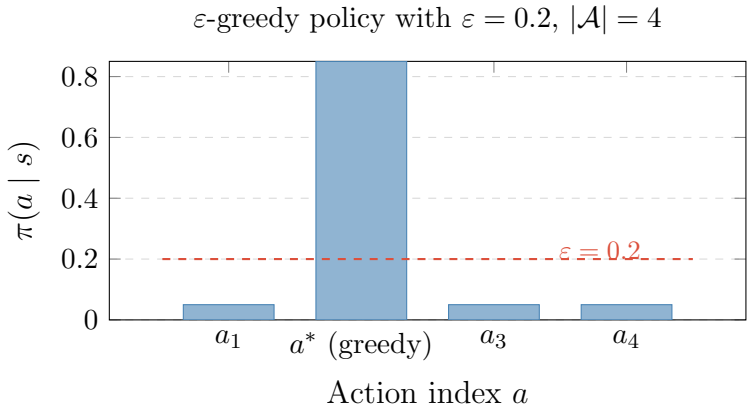



Figure 3: Action probabilities under an $\varepsilon$-greedy policy with $\varepsilon = 0.2$ and $|\mathcal{A}| = 4$. The greedy action $a^*$ receives probability 0.85; all others receive $\varepsilon/|\mathcal{A}| = 0.05$.

## 6.3 On-Policy MC Control with $\varepsilon$-Greedy

**Algorithm 3** On-Policy MC Control with $\varepsilon$-Greedy Exploration

**Initialise:** $Q(s, a)$ arbitrarily; $\pi(a|s) \leftarrow \varepsilon$-greedy w.r.t. $Q$
**repeat**
  Generate episode $S_0, A_0, R_1, \ldots, S_T$ following $\pi$
  $G \leftarrow 0$
  **for** $t = T-1, \ldots, 0$ **do**
    $G \leftarrow \gamma G + R_{t+1}$
    **if** $(S_t, A_t)$ first visit in this episode **then**
      $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha[G - Q(S_t, A_t)]$
      Update $\pi(\cdot|S_t)$ to be $\varepsilon$-greedy w.r.t. $Q(S_t, \cdot)$
    **end if**
  **end for**
**until** convergence

## 6.4 Convergence and the Best $\varepsilon$-Soft Policy

**Theorem 6.1** (Convergence of On-Policy MC)**.** *Any $\varepsilon$-greedy policy with respect to the optimal $Q$ function among all $\varepsilon$-soft policies $q^*$ is an $\varepsilon$-soft policy. On-policy first-visit MC control converges to the optimal $\varepsilon$-soft policy, i.e., the policy that maximises $V_\pi(s)$ subject to being $\varepsilon$-soft.*

The optimal $\varepsilon$-soft policy differs from the true optimal $\pi^*$ by at most:

$$V_{\pi^*}(s) - V_{\pi^*_\varepsilon}(s) \;\leq\; \frac{\varepsilon}{1-\gamma} \cdot \max_{s,a} |Q^*(s,a)|.$$

As $\varepsilon \to 0$, the bound vanishes and we recover the optimal policy—but at the cost of less exploration. In practice, $\varepsilon$ is often annealed (gradually reduced) over training.

### 6.5 Exploration-Exploitation Trade-off

**Trade-off Summary**

**High $\varepsilon$:** More exploration $\Rightarrow$ better coverage of $(s,a)$ space $\Rightarrow$ more accurate $Q$ estimates, but slower convergence to a good policy.

**Low $\varepsilon$:** More exploitation $\Rightarrow$ policy focuses on promising actions $\Rightarrow$ faster reward accumulation, but risk of missing better alternatives.

**Annealing:** A common heuristic: start with high $\varepsilon$ (explore broadly) and decay toward 0 (exploit learned knowledge).

## 7. Off-Policy Monte Carlo Prediction

### 7.1 Motivation

On-policy MC learns the value of the *same* policy it uses to generate data. This creates a fundamental tension: a good exploration policy (stochastic, wide-ranging) is not the same as a good exploitation policy (greedy, focused). **Off-policy methods** resolve this by separating the two roles:

- **Target policy** $\pi$: the policy whose value we want to learn (typically deterministic and near-optimal).
- **Behaviour policy** $b$: the policy used to generate episodes (exploratory, stochastic). Must satisfy **coverage**: $\pi(a|s) > 0 \Rightarrow b(a|s) > 0$.

Off-policy learning is powerful because it can learn from human demonstrations, historical data, or a separate exploration agent while still targeting the optimal deterministic policy.

### 7.2 Importance Sampling

The challenge is that episodes are generated under $b$, but we want to estimate expectations under $\pi$. Importance sampling corrects for this distributional mismatch.

**7.2.1 Derivation of the Importance Sampling Ratio.** Consider an episode generated from time $t$ to termination $T$. The probability of the observed trajectory $\tau = (S_t, A_t, S_{t+1}, A_{t+1}, \ldots, S_T)$ under policy $\mu$ is:

$$P(\tau \mid \mu) = \prod_{k=t}^{T-1} \mu(A_k \mid S_k)\, \mathcal{P}(S_{k+1} \mid S_k, A_k).$$

The **importance sampling ratio** is the likelihood ratio between $\pi$ and $b$:

$$\rho_{t:T} = \frac{P(\tau \mid \pi)}{P(\tau \mid b)} = \frac{\prod_{k=t}^{T-1} \pi(A_k \mid S_k)\, \mathcal{P}(S_{k+1} \mid S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k \mid S_k)\, \mathcal{P}(S_{k+1} \mid S_k, A_k)}$$

$$= \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}. \tag{10}$$

The transition probabilities $\mathcal{P}$ cancel because they are the same under both policies—the environment is fixed. This is a crucial simplification: we need only the policy probabilities, not the model.

$$\boxed{\rho_{t:T} = \prod_{k=t}^{T-1} \frac{\pi(A_k \mid S_k)}{b(A_k \mid S_k)}}$$

The ratio $\rho_{t:T}$ is large when actions taken under $b$ were more likely under $\pi$, and small when $b$ explored actions that $\pi$ would rarely take.

## 7.3 Ordinary vs Weighted Importance Sampling

Given $n$ episodes, let $\mathcal{J}(s)$ denote the set of time steps at which state $s$ is visited (first-visit or every-visit), and let $T(j)$ be the termination time of the episode containing step $j$.

### 7.3.1 Ordinary (Standard) Importance Sampling.

$$\widehat{V}^{\text{OIS}}(s) = \frac{\sum_{j \in \mathcal{J}(s)} \rho_{j:T(j)}\, G_j}{|\mathcal{J}(s)|}. \tag{11}$$

**Properties:** Unbiased estimator of $V_\pi(s)$, but can have very high or even infinite variance. A single trajectory where $\rho \gg 1$ can dominate the estimate.

### 7.3.2 Weighted Importance Sampling.

$$\widehat{V}^{\text{WIS}}(s) = \frac{\sum_{j \in \mathcal{J}(s)} \rho_{j:T(j)}\, G_j}{\sum_{j \in \mathcal{J}(s)} \rho_{j:T(j)}}. \tag{12}$$

**Properties:** Biased for finite samples (but consistent), and has *substantially lower variance* in practice. WIS is the preferred method in most applications.

## 7.4 Bias-Variance Trade-off

To see why OIS variance can be infinite: if $\rho_{t:T}$ has heavy tails (which occurs when $b$ sometimes assigns very low probability to actions that $\pi$ always takes), then $\text{Var}[\rho G]$ may diverge.

Table 3: Ordinary vs Weighted Importance Sampling

| **Property** | **Ordinary IS** | **Weighted IS** |
|---|---|---|
| Bias (finite $n$) | Unbiased | Biased |
| Bias ($n \to \infty$) | Unbiased | Consistent (bias $\to 0$) |
| Variance | Can be infinite | Bounded; much lower |
| Stability | Prone to outliers | Robust to large $\rho$ |
| Preferred? | Rarely in practice | Yes, in most settings |

**7.4.1 Incremental Implementation of WIS.** Weighted importance sampling can be maintained incrementally. Let $C_n(s,a)$ be the cumulative sum of importance ratios after $n$ updates. The update rule is:

$$C_n(s,a) \leftarrow C_{n-1}(s,a) + W_n, \tag{13}$$

$$Q_n(s,a) \leftarrow Q_{n-1}(s,a) + \frac{W_n}{C_n(s,a)}\big[G - Q_{n-1}(s,a)\big], \tag{14}$$

where $W_n = \rho_{t:T(j)}$ is the importance ratio for the $n$-th visit.

**Algorithm 4** Off-Policy MC Prediction (Weighted Importance Sampling)

**Input:** target policy $\pi$, behaviour policy $b$
**Initialise:** $Q(s,a) \leftarrow 0$; $C(s,a) \leftarrow 0$ for all $(s,a)$
**repeat**
  Generate episode $S_0, A_0, R_1, \ldots, S_T$ following $b$
  $G \leftarrow 0$; $W \leftarrow 1$
  **for** $t = T{-}1, T{-}2, \ldots, 0$ **do**
    $G \leftarrow \gamma G + R_{t+1}$
    $C(S_t, A_t) \leftarrow C(S_t, A_t) + W$
    $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \dfrac{W}{C(S_t, A_t)}\big[G - Q(S_t, A_t)\big]$
    $W \leftarrow W \cdot \dfrac{\pi(A_t \mid S_t)}{b(A_t \mid S_t)}$
    **if** $W = 0$ **then break** ▷ $\pi$ would never take $A_t$; no further updates
    **end if**
  **end for**
**until** convergence

## 7.5 Coverage Requirement

Off-policy methods require the **coverage assumption**:

$$\pi(a \mid s) > 0 \;\Rightarrow\; b(a \mid s) > 0, \qquad \forall (s,a).$$

If $b$ assigns zero probability to an action that $\pi$ would take, the ratio $\pi/b$ is undefined, and the algorithm cannot correct for this. A simple sufficient choice is $b =$ uniform random policy over $\mathcal{A}$, which trivially satisfies coverage for any deterministic target $\pi$.

# 8. Summary and Discussion

## 8.1 Unified View of MC Methods

Monte Carlo methods form a coherent family of model-free algorithms distinguished by their use of complete episode returns rather than bootstrapped estimates. Table 4 summarises the key algorithms discussed in this report.

Table 4: Summary of Monte Carlo algorithms covered

| **Algorithm** | **On/Off policy** | **Exploration** | **Key feature** |
|---|---|---|---|
| MC Prediction (state values) | On-policy | N/A | Estimates $V_\pi$ |
| MC Prediction (action values) | On-policy | N/A | Estimates $Q_\pi$ |
| MC-ES | On-policy | Exploring starts | Exact convergence |
| On-policy MC ($\varepsilon$-greedy) | On-policy | $\varepsilon$-greedy | Practical; converges to $\varepsilon$-optimal |
| Off-policy MC (IS) | Off-policy | Via $b$ | Learns $\pi$ from $b$'s data |

## 8.2 Strengths of Monte Carlo Methods

1. **Model-free**: No knowledge of $\mathcal{P}$ or $\mathcal{R}$ required.
2. **No bootstrapping**: Estimates are based on actual returns, not noisy value estimates, avoiding the propagation of bias.
3. **Efficient in complex MDPs**: Updates affect only visited states, making MC computationally efficient when only a small portion of the state space is relevant.
4. **Applicable to non-Markovian environments**: Since MC uses full trajectories, it can (with care) be applied even when the Markov property is not strictly satisfied.

## 8.3 Limitations

1. **Episodic requirement**: Standard MC requires episodes to terminate. Continuing tasks require modifications (e.g., truncated returns).
2. **High variance**: Using the full return $G_t$ introduces high variance, particularly in long episodes with large $\gamma$.
3. **Slow learning**: Since updates occur only after episode termination, MC can be slower than temporal-difference methods that update after every step.
4. **Importance sampling instability**: Off-policy MC with OIS can have unstable estimates, especially when $\pi$ and $b$ are very different.

### 8.4 Connection to Temporal-Difference Methods

MC methods lie at one extreme of a spectrum parameterised by the $n$**-step return**:

$$G_t^{(n)} = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{n-1} R_{t+n} + \gamma^n V(S_{t+n}).$$

At $n = 1$ this is the TD(0) update; at $n = T - t$ (full episode) this is the MC return. The family of $n$-step TD methods interpolates between these extremes, and eligibility traces (TD($\lambda$)) provide an efficient way to blend them. Understanding MC methods is therefore foundational to understanding the broader landscape of RL algorithms.

## 9. Conclusion

This report has presented a comprehensive treatment of Monte Carlo methods in reinforcement learning. We began with the fundamental problem of policy evaluation, introducing first-visit and every-visit MC prediction and the incremental update rule. We argued that action-value estimation is necessary for model-free control, and derived the MC backup diagram showing that no bootstrapping is used.

For control, we studied MC-ES, which elegantly guarantees convergence via exploring starts, and identified its practical limitations. We then introduced $\varepsilon$-greedy on-policy methods as a robust alternative that sacrifices a small degree of optimality for practical applicability.

Finally, we derived the importance sampling ratio from first principles and contrasted ordinary and weighted importance sampling, showing that weighted IS offers substantially lower variance at the cost of a small finite-sample bias—making it the method of choice in practice.

Monte Carlo methods remain a fundamental and practically important tool in RL. Their simplicity, model-free nature, and lack of bootstrapping make them indispensable, particularly in domains where episodes are well-defined and the full return is observable.

# A. Mathematical Proofs and Derivations

## A.1 Unbiasedness of First-Visit MC

*Proof.* Let $\tau_1, \tau_2, \ldots, \tau_n$ be independent episodes. For state $s$, let $G^{(i)}$ be the return from the first visit to $s$ in episode $i$ (if it occurs). Since episodes are independent, the returns $G^{(1)}, G^{(2)}, \ldots$ are i.i.d. with expectation $\mathbb{E}[G^{(i)}] = V_\pi(s)$. By the strong law of large numbers:

$$\widehat{V}(s) = \frac{1}{n}\sum_{i=1}^{n} G^{(i)} \xrightarrow{\text{a.s.}} \mathbb{E}[G^{(i)}] = V_\pi(s).$$

□

## A.2 Cancellation of Transition Probabilities in IS Ratio

The key step in deriving (10) is:

$$\rho_{t:T} = \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k)\, \mathcal{P}(S_{k+1}|S_k, A_k)}{\prod_{k=t}^{T-1} b(A_k|S_k)\, \mathcal{P}(S_{k+1}|S_k, A_k)} = \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)}.$$

This cancellation means $\rho_{t:T}$ can be computed without knowing $\mathcal{P}$—a crucial property that makes off-policy MC truly model-free.

## A.3 Variance of Ordinary Importance Sampling

Let $X = \rho_{t:T} G_t$. The variance of the OIS estimator scales with $\mathrm{Var}[X]$. Since:

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[\rho_{t:T}^2 G_t^2] - (V_\pi(s))^2,$$

the term $\mathbb{E}[\rho_{t:T}^2 G_t^2]$ can be infinite if the tails of $\rho$ are heavy. This occurs, for example, when $b(a|s) \ll \pi(a|s)$ for some high-reward actions—the ratio $\pi/b$ can be arbitrarily large.

Weighted IS avoids this by normalising: the ratio of sums has bounded range $[0, \max_j G_j]$, guaranteeing finite variance.

# B. Notation Reference

Table 5: Key notation used throughout the report

| Symbol | Meaning |
|---|---|
| $\mathcal{S}, \mathcal{A}$ | State and action spaces |
| $\mathcal{P}(s'\|s,a)$ | Transition probability |
| $\mathcal{R}(s,a,s')$ | Expected reward |
| $\gamma \in [0,1)$ | Discount factor |
| $\pi(a\|s)$ | Policy: probability of action $a$ in state $s$ |
| $V_\pi(s)$ | State-value function under $\pi$ |
| $Q_\pi(s,a)$ | Action-value function under $\pi$ |
| $G_t$ | Discounted return from time $t$ |
| $T$ | Terminal time step of an episode |
| $N(s)$ | Number of visits to state $s$ |
| $\alpha$ | Step-size (learning rate) |
| $\varepsilon$ | Exploration parameter |
| $\rho_{t:T}$ | Importance sampling ratio from $t$ to $T$ |
| $b$ | Behaviour policy (off-policy methods) |

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